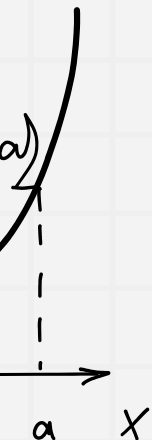


$$\int_0^1 dy \int_0^{1-x} f(x) dx + \int_0^{1/\sqrt{2}} dy \int_0^{f(2x)} f(2x) dx =$$

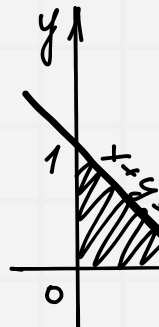
$$2\sqrt{y^2 - x^2}$$



Números Binários

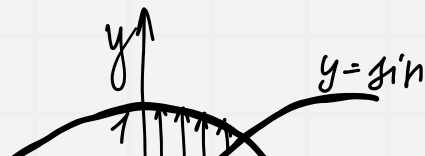
Matemática Computacional Aplicada

Daniel Henrique Matos de Paiva



$$\begin{aligned} X &= 2y \\ Z &= 1 + y \\ Z &= 4 + y \end{aligned}$$

$$\int_0^1 dx \int_0^{1-x} x^2 z^{10(x+3y)} du =$$



$$\int_0^{1-x} \int_0^{10(x+3y)} \int_0^{10(x+3y)} x^2 dz =$$

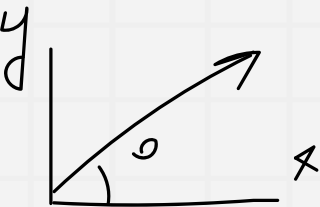
$$\frac{3, x=5}{2+4y^2} \cdot \frac{1}{1+4y^2}$$

$$V: z=10(x+3y), x+y+z=10, x=0, y=0, z=0$$

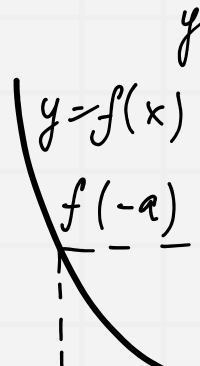
01

Adição

Como funciona?



$$= \int_0^1$$



fdy

Regras

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$1 + 1 = 0$ (e "vai 1" para o dígito de ordem superior)

$1 + 1 + 1 = 1$ (e "vai 1" para o dígito de ordem superior)



$$5 = 2$$

^r
f dy

Regras

$$101_{(2)} + 011_{(2)} = ?$$



$$5 = 2$$

1

1

1

1 0 1 (bin)

+ 0 1 1 (bin)

1 0 0 0 (bin)

^r
f dy

Regras

$$11010_{(2)} + 10100_{(2)} = ?$$



$$5 = 2$$

1

1 1 0 1 0 (bin)

+ 1 0 1 0 0 (bin)

1 0 1 1 1 0 (bin)

fdy

Regras

11011011₍₂₎ +

11110110₍₂₎ = ?



5 = 2

	1	1	1	1	1	1	1		
	1	1	0	1	1	0	1	1	(bin)
+	1	1	1	1	0	1	1	0	(bin)
1	1	1	0	1	0	0	0	1	(bin)



A hand-drawn diagram illustrating a 4x4 grid. The grid is represented by vertical lines. A 2x2 subgrid in the top-left corner is highlighted with a red oval. An arrow points to the right side of the grid. Below the grid, a horizontal line separates it from the text '1 0 0 1 (2)'.

$$\begin{array}{r} \downarrow \quad 3 \\ + \quad 3 \\ \hline (2) \quad 9_{(10)} \end{array}$$

$$\begin{array}{r}
 1 \quad 1 \quad 1 \quad 0 \quad 0 \\
 + \quad 1 \quad 1 \quad 0 \\
 + \quad 1 \quad 1 \quad 1 \\
 \hline
 1 \quad 0 \quad 0 \quad 0 \quad 1_{(2)}
 \end{array}$$

$$\begin{array}{r}
 \overset{1}{1} \overset{1}{0} \overset{1}{0} \overset{1}{1} \\
 + \quad 1 \ 1 \ 1 \ 1 \ 0 \\
 \hline
 1 \ 1 \ 0 \ 0 \ 0 \ 1 \text{ (2)}
 \end{array}$$

Handwritten binary addition:

$$\begin{array}{r}
 1001 \\
 + 1101 \\
 \hline
 100101_{(2)}
 \end{array}$$

The first row is 1001, the second row is + 1101, and the result is 100101₍₂₎. A pink oval highlights the first four digits of the first row (1001). The result 100101 is written in green.